

Higher Diet Quality Is Associated with Decreased Risk of All-Cause, Cardiovascular Disease, and Cancer Mortality among Older Adults



Increased attention in dietary research and guidance has been focused on dietary patterns, rather than on single nutrients or food groups, because dietary components are consumed in combination and correlated with one another. However, the collective body of research on the topic has been hampered by the lack of consistency in methods used. We examined the relationships between 4 indices—the Healthy Eating Index–2010 (HEI-2010), the Alternative Healthy Eating Index–2010 (AHEI-2010), the alternate Mediterranean Diet (aMED), and Dietary Approaches to Stop Hypertension (DASH)—and all-cause, cardiovascular disease (CVD), and cancer mortality in the NIH-AARP Diet and Health Study (n = 492,823). Data from a 124-item food-frequency questionnaire were used to calculate scores; adjusted HRs and 95% CIs were estimated. We documented 86,419 deaths, including 23,502 CVD- and 29,415 cancer-specific deaths, during 15 y of follow-up. Higher index scores were associated with a 12–28% decreased risk of all-cause, CVD, and cancer mortality. Specifically, comparing the highest with the lowest quintile scores, adjusted HRs for all-cause mortality for men were as follows: HEI-2010 HR: 0.78 (95% CI: 0.76, 0.80), AHEI-2010 HR: 0.76 (95% CI: 0.74, 0.78), aMED HR: 0.77 (95% CI: 0.75, 0.79), and DASH HR: 0.83 (95% CI: 0.80, 0.85); for women, these were HEI-2010 HR: 0.77 (95% CI: 0.74, 0.80), AHEI-2010 HR: 0.76 (95% CI: 0.74, 0.79), aMED HR: 0.76 (95% CI: 0.73, 0.79), and DASH HR: 0.78 (95% CI: 0.75, 0.81). Similarly, high adherence on each index was protective for CVD and cancer mortality examined separately. These findings indicate that multiple scores reflect core tenets of a healthy diet that may lower the risk of mortality outcomes, including federal guidance as operationalized in the HEI-2010, Harvard’s Healthy Eating Plate as captured in the AHEI-2010, a Mediterranean diet as adapted in an Americanized aMED, and the DASH Eating Plan as included in the DASH score.